

ITU-T (G.8013) Y.1731

OAM functions and mechanisms for Ethernet-based networks

Revision history:

Revision	Author	Contact Info	Description	Date
0.1	Yamin		Learning report of OAM functions and mechanisms for Ethernet based networks.	

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1. Introduction

OAM Functions and Mechanisms for Ethernet-based Networks, ITU-T standard Y.1731, is one of the many building blocks for Ethernet OAM. (Other OAM standards include Institute of Electrical and Electronics Engineers (IEEE) 802.1ag for connectivity fault management, IEEE 802.3ah for Ethernet link OAM, and Metro Ethernet Forum (MEF) Ethernet Local Management Interface (ELMI).

Y.1731 is a service layer protocol that measures service-level agreement parameters end-to-end from user-to-network interface (UNI) to UNI. To measure service-level agreement parameters such as delay or frame delay variation, a small number of synthetic frames are transmitted along with the service to the end point of the maintenance region, where the MEP responds to the synthetic frame.

The following procedures are included in the ITU-T Y.1731 standard:

- ✧ Connectivity fault management (CFM)
- ✧ Diagnostics (including throughput)
- ✧ Threshold-crossing events for out-of-bounds measurements
- ✧ Performance management
 - Frame delay
 - Frame delay variation (jitter)
 - Frame loss

1.1 Maintenance Entity Group (MEG)

A maintenance entity group (MEG) object, as defined by Y.1731, includes different maintenance entities (MEs) that satisfy the following conditions:

- ✓ MEs in a MEG exist in the same administrative boundary.
- ✓ MEs in a MEG have the same MEG level.
- ✓ MEs in a MEG belong to the same point-to-point or multipoint Ethernet connection.

A MEG is a collection of MEs, and an ME refers to the direct connection between two MEPs in a MEG.

- ✓ For a point-to-point ETH connection, a MEG contains a single ME.
- ✓ For a multipoint ETH connection containing n end points, a MEG contains $n \times (n - 1) / 2$ MEs

A MEP forms an ME with every other MEP in the same MEG. A MEP has the ability to check for continuity and measure network performance across any of the MEs in its MEG where it is one of the two end-points. Additionally, the MEG's level attribute is used by all MEPs created within it.

The minimum attributes required to create or access a MEG are the MEG ID and MEG level after which the MEG attributes can be edited to make the MEG ready for operation. The reason why both MEG ID and MEG level must be used to create the MEG is to allow MEGs with the same name but different MEG levels to be unique. This may be required if a device must interoperate

with other devices not under the control of the administrator that use the same MEG ID but operate at different MEG levels.

1.2 MEG Level

The MEG level is an attribute of a MEG that allows multiple Y.1731 services to use the same MEG name on a link but remain independent from one another if each service is using a different level. The MEG level range is 0 to 7 and the MEG level configured for a particular MEG is embedded in all Y.1731 frames transmitted by its MEPs.

The MEG level attribute is useful when Y.1731 services are nested to allow higher levels to operate across larger areas of the network compared to lower levels. Generally, a MEP will process all Y.1731 frames with a MEG level equal to or lower than its own and on the same service. Frames with a higher MEG level will not be processed, and the lower level MEP will be transparent resulting in the higher level frames being forwarded as normal data traffic

Eight MEG levels are available to accommodate different network deployment scenarios.

When customer, provider and operator data path flows are not distinguishable based on the means of the ETH layer encapsulations, the eight MEG levels can be shared amongst them to distinguish between OAM frames belonging to nested MEGs of customers, providers and operators.

The default MEG level assignment amongst customer, provider and operator roles is:

- ✓ The customer role is assigned three MEG levels: 7, 6 and 5.
- ✓ The provider role is assigned two MEG levels: 4 and 3.
- ✓ The operator role is assigned three MEG levels: 2, 1 and 0.

The default MEG level assignment can be changed via a mutual agreement among customer, provider and/or operator roles.

1.3 MEG ID

The MEG ID is composed of the MEG name format and MEG name. On a network device, the MEG ID must be unique on a per-MEG level basis. Two MEGs at the same MEG level cannot have the same MEG ID. By combining the MEG ID within the MEG level, multiple MEGs on the same device can use the same MEG ID if they exist at different MEG levels. An example is shown in Table 1. MEG-A in MEG level 2 and MEG-A in MEG level 4 are unique as they are on different MEG levels. Their services can be uniquely identified when parsing a Y.1731 frame by combining the MEG level and MEG ID values. The two MEG-D MEGs in MEG level 6 are not unique because they are operating on the same MEG level; therefore, their frames cannot be uniquely distinguished between one another.

Table 1. MEG ID Example

MEG level	MEG IDs
2	MEG-A, MEG-B
4	MEG-A, MEG-C
6	MEG-D, MEG-D

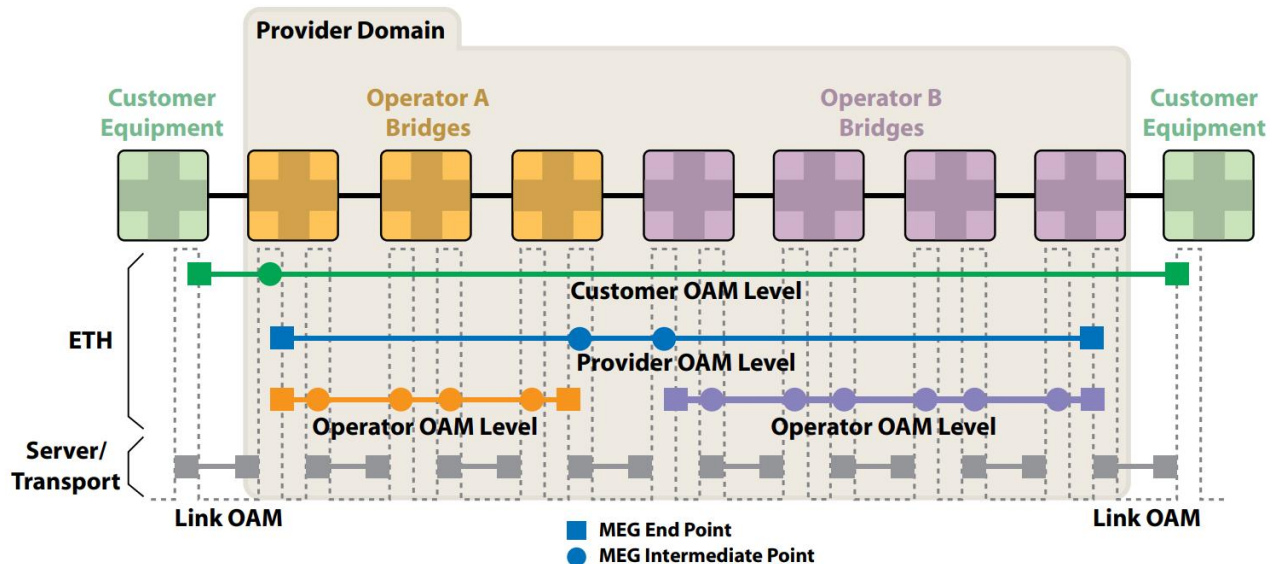
The MEG name format identifies the format of the MEG name and is present in Y.1731 frames transmitted by MEPs. The name formats supported are character strings and the ITU-T Carrier Code (ICC)-based format. The use of character strings allows MEPs in the MEG to interoperate with MEPs created on devices supporting 802.1ag.

The character string format indicates the name is composed of ASCII characters and the valid range for its length is 1 to 45 characters.

The ICC-based format consists of 13 characters of two subfields: the ICC followed by a unique MEG code (UMC).

1.4 Multi-Domain Network Model

The multi-domain network model in ITU-T Y.1731 defines how Ethernet networks spanning multiple administrative domains are managed and monitored. These domains are divided into **Customer**, **Provider**, and **Operator** maintenance levels, each responsible for specific OAM tasks.



- **Customer OAM Level:** Focuses on end-to-end service monitoring from the customer's perspective, with MEPs located at customer equipment boundaries.
- **Provider OAM Level:** The provider ensures complete service delivery across their infrastructure, with MEPs at boundaries and MIPs within the network for fault management.
- **Operator OAM Level:** Operators handle transport across subnetworks, with MIPs on bridges or transport devices to support the provider's service.

A **Maintenance Entity Group (MEG)** includes a set of **Maintenance Entities (MEs)** within the same domain and level. Each ME represents the relationship between two **MEPs**. In point-to-point connections, a MEG contains a single ME, while in multipoint connections, it can have multiple MEs, depending on the endpoints.

This model supports hierarchical OAM with **eight MEG levels (0 to 7)**, ensuring distinct responsibilities and transparent OAM flows between domains, facilitating fault management and performance monitoring across Ethernet networks.

2. Performance monitoring

ITU-T Y.1731 performance monitoring provides standards-based Ethernet performance monitoring that encompasses the measurement of Ethernet frame delay, frame delay variation, and frame loss and throughput as outlined in the ITU-T Y-1731 specification and interpreted by the Metro Ethernet Forum (MEF). Service providers offer service-level agreements (SLAs) that describe the level of performance customers can expect for services.

2.1 Ethernet Frame Loss Measurement (ETH-LM)

Ethernet frame loss measurement (ETH-LM) enables a local MEP and its RMEP to exchange ETH-LM packets to collect frame loss statistics on E2E links. ETH-LM has two modes: near- and far-end ETH-LM.

Near-end ETH-LM applies to an inbound interface on a MEP, and far-end ETH-LM applies to the outbound interface. ETH-LM counts the number of faulty frame seconds to determine the duration in which a link is unavailable.

2.1.1 Single-Ended Frame Loss Measurement

Single-ended frame loss measurement in ITU-T Y.1731 can be performed in two modes: on-demand or proactive.

1. **On-Demand Measurement:** Manually initiated to diagnose frame loss over a specific time period. Can be executed once or periodically, depending on the diagnostic requirement.

2. **Proactive Measurement:** Continuously monitors frame loss to enable proactive detection of service degradation. Helps maintain ongoing visibility into network performance.

Both methods operate by exchanging ETH-LM messages between a local MEP (Maintenance End Point) and a remote MEP (RMEP).

Message Types

- 1) **LMM (Loss Measurement Message):** Contains the frame loss measurement request from the local MEP.
- 2) **LMR (Loss Measurement Reply):** Contains the response from the RMEP.

Procedure

1. After enabling single-ended frame loss measurement, the local MEP (e.g., on PE1) sends an ETH-LMM carrying a request to the RMEP (e.g., on PE2).
2. The RMEP responds with an ETH-LMR, which includes the following information:
 - ✓ **TxFcf:** The value of the local transmit counter when the LMM was sent (copied from the received ETH-LMM).
 - ✓ **RxFcf:** The value of the local receive counter when the ETH-LMM was received.
 - ✓ **TxFcb:** The value of the local transmit counter when the ETH-LMR was sent.
3. Upon receiving the ETH-LMR, the initiating MEP calculates the near-end and far-end frame loss using the following data:
 - ✧ **Current ETH-LMR Values:**
 - ✓ **TxFcf[tc]:** Transmit counter value from the ETH-LMM.
 - ✓ **RxFcf[tc]:** Receive counter value when the ETH-LMM was received.
 - ✓ **TxFcb[tc]:** Transmit counter value when the ETH-LMR was sent.
 - ✓ **RxFcl[tc]:** Local receive counter value when the ETH-LMR was received.
 - ✧ **Previous ETH-LMR Values:**
 - ✓ **TxFcf[tp], RxFcf[tp], TxFcb[tp], and RxFcl[tp]** represent similar counter values from the previously received ETH-LMR.

Frame Loss Calculation

The near-end and far-end frame losses are calculated using the difference between consecutive counter values from the ETH-LMR frames:

$$\text{Frame Loss}_{\text{far-end}} = |\text{TxFcf}[\text{tc}] - \text{TxFcf}[\text{tp}]| - |\text{RxFcf}[\text{tc}] - \text{RxFcf}[\text{tp}]|$$

$$\text{Frame Loss}_{\text{near-end}} = |\text{TxFcb}[\text{tc}] - \text{TxFcb}[\text{tp}]| - |\text{RxFcl}[\text{tc}] - \text{RxFcl}[\text{tp}]|$$

This process ensures precise measurement of frame loss in both directions, helping network operators monitor and maintain service quality effectively.

2.1.2 Dual-Ended Frame Loss Measurement

Dual-ended frame loss measurement is a proactive method that periodically measures frame loss between two Maintenance End Points (MEPs), supporting error management and performance monitoring. Unlike single-ended measurement, both MEPs in this method actively send and process ETH-LM messages. The received messages are not forwarded, ensuring localized loss statistics.

Procedure

Each MEP (e.g., on PE1 and PE2) periodically transmits an ETH-LM message to its corresponding Remote MEP (RMEP). The Continuity Check Message (CCM) carries the required counter values for frame loss calculation.

Data in the CCM:

Each CCM contains the following information:

- ✓ TxFCf: The value of the local transmit counter when the CCM is sent.
- ✓ RxFCb: The value of the local receive counter from the last received CCM.
- ✓ TxFCb: The value of the transmit counter in the last received CCM.

Frame Loss Calculation:

Upon receiving a CCM, the MEP processes the received information and calculates the near-end and far-end frame losses using the following:

- ✧ Current CCM Values:
 - ✓ TxFCf[tc]: Transmit counter value from the current CCM.
 - ✓ RxFCb[tc]: Receive counter value from the last CCM.
 - ✓ TxFCb[tc]: Transmit counter value in the last received CCM.
 - ✓ RxFCI[tc]: Local receive counter value at the time of receiving the current CCM.
- ✧ Previous CCM Values:
 - ✓ TxFCf[tp], RxFCb[tp], TxFCb[tp], and RxFCI[tp], from the previously received CCM.

Frame Loss Formulas:

$$\text{Frame Loss}_{\text{far-end}} = |\text{TxFCb}[\text{tc}] - \text{TxFCb}[\text{tp}]| - |\text{RxFCb}[\text{tc}] - \text{RxFCb}[\text{tp}]|$$

$$\text{Frame Loss}_{\text{near-end}} = |\text{TxFCf}[\text{tc}] - \text{TxFCf}[\text{tp}]| - |\text{RxFCI}[\text{tc}] - \text{RxFCI}[\text{tp}]|$$

Dual-ended frame loss measurement is essential for carrier-grade networks, allowing service providers to maintain high-quality Ethernet services across complex networks.

2.2 Ethernet Delay Measurement (ETH-DM)

The ETH-DM (Ethernet Delay Measurement) function in ITU-T Y.1731 measures frame transmission delays and their variations by exchanging delay measurement messages between a Maintenance End Point (MEP) and its Remote MEP (RMEP). This mechanism enables operators to monitor the performance of Ethernet services in terms of latency and jitter.

2.2.1 One-Way Frame Delay Measurement

One-way delay measurement calculates the time taken for a frame to travel from a MEP to its RMEP. It requires the MEP to send a 1DM (One-Way Delay Measurement Message) to the RMEP, which then computes the delay using timestamp values.

Requirements

- ✓ The MEP and RMEP must synchronize their clocks to measure one-way frame delay accurately.
- ✓ If synchronization is unavailable, only delay variation (jitter) can be measured, as it does not rely on synchronized clocks.

Measurement Modes

1. On-Demand One-Way Delay Measurement:
 - Manually initiated to diagnose delays over a specific time frame.
 - Can be performed as a one-time or periodic test.
2. Proactive One-Way Delay Measurement:
 - Continuously monitors delays to provide real-time performance insights and proactive reporting.

Procedure

- ✧ After configuration, the MEP periodically sends 1DM packets to its RMEP.
- ✧ Each 1DM packet contains TxTimeStampf, the timestamp indicating when the packet was transmitted.
- ✧ Upon receiving the 1DM, the RMEP compares TxTimeStampf with RxTimef (the time when the packet was received) to calculate the delay:

$$\text{Frame Delay} = \text{RxTimef} - \text{TxTimeStampf}$$

Delay variation is calculated as the absolute difference between two consecutive frame delay values:

$$\text{Delay Variation} = | \text{Frame Delay}_n - \text{Frame Delay}_{n-1} |$$

One-way frame delay measurement in Y.1731, using 1DM messages, provides precise latency monitoring when time synchronization is available. It also supports proactive and on-demand performance

monitoring, ensuring network operators can maintain high-quality Ethernet services with accurate, priority-based delay statistics.

2.2.2 Two-Way Frame Delay Measurement

Two-way frame delay measurement is a method in ITU-T Y.1731 used to calculate the round-trip delay between a Maintenance End Point (MEP) and its Remote MEP (RMEP) by exchanging Delay Measurement Messages (DMMs) and Delay Measurement Replies (DMRs). This method does not require clock synchronization between the MEP and RMEP, making it widely applicable for performance monitoring.

Modes of Measurement

1. On-Demand Measurement:

- Performed manually to diagnose frame delay at specific times.
- Can be executed as a one-time or periodic test.

2. Proactive Measurement:

- Performed continuously to monitor round-trip delays in real time.
- Enables proactive identification of performance issues.

Procedure

1. After two-way frame delay measurement is configured, the local MEP periodically sends DMMs to its RMEP.
2. Each DMM contains:
 - ✓ TxTimeStamp: Timestamp indicating when the DMM was sent by the MEP.
3. Upon receiving the DMM, the RMEP replies with a DMR, which includes:
 - ✓ RxTimeStamp: Timestamp indicating when the DMM was received.
 - ✓ TxTimeStamp: Timestamp indicating when the DMR was sent.
 - ✓ All other fields from the DMM are copied into the DMR, except the source and destination MAC addresses, which are swapped.
4. When the local MEP receives the DMR, it calculates the two-way frame delay using the formula:

$$\text{Frame Delay} = (\text{RxTimeb} - \text{TxTimeStampf}) - (\text{TxTimeStampb} - \text{RxTimeStampf})$$

5. The frame delay variation (jitter) is calculated as the absolute difference between two consecutive frame delay values:

$$\text{Delay Variation} = |\text{Frame Delay}_n - \text{Frame Delay}_{n-1}|$$

Two-way frame delay measurement is a critical tool for ensuring high-quality Ethernet service delivery in carrier-grade networks.

2.3 Throughput Measurement

Throughput measurement determines the maximum rate at which a network link can transmit data without any packet loss. This is essential for understanding the performance limits of the network and identifying potential bottlenecks. ITU-T Y.1731 supports throughput measurement using mechanisms such as **unicast ETH-LB (Loopback)** and **ETH-Test** functions.

Mechanisms for Throughput Measurement

- ✧ ETH-LB: Sends LBM (Loopback Message) and receives LBR (Loopback Reply) to test one-way or two-way throughput by increasing transmission rates until frame loss occurs.
- ✧ ETH-Test: Tests throughput, frame loss, and bit errors using TST frames with configurable data sizes and patterns. Verifies throughput by reducing the transmission rate until no packet loss occurs. Detects bit errors by comparing CRC values between the sender and receiver.

2.4 Ethernet Synthetic Loss Measurement (ETH-SLM)

ETH-SLM is a mechanism in ITU-T Y.1731 used to measure frame loss by transmitting synthetic frames rather than relying on actual data traffic. This method provides a statistical sample that approximates the frame loss ratio in a network. The measurement is conducted by exchanging ETH-SLM frames between Maintenance End Points (MEPs) to track the number of transmitted and received synthetic frames.

Functionality

- ✓ ETH-SLM can be performed in on-demand or proactive mode:
 - On-demand mode: Manually initiated to assess network performance over a limited period.
 - Proactive mode: Continuously monitors frame loss for ongoing performance evaluation.
- ✓ Each MEP maintains counters to track transmitted (TxFCI) and received (RxFCI) synthetic frames for each monitored peer MEP.
- ✓ Since a bidirectional service is considered unavailable if frame loss occurs in either direction, ETH-SLM ensures that near-end and far-end loss measurements contribute to identifying service availability.

Configuration Parameters

A MEP requires the following configurations for ETH-SLM:

- ✓ MEG level: Defines the hierarchical level of the measurement.
- ✓ Destination MAC address: Identifies the peer MEP receiving the synthetic frames.
- ✓ Test ID: Uniquely identifies each synthetic loss measurement session.
- ✓ Priority: Configurable per operation to define the priority of ETH-SLM frames.

- ✓ Drop eligibility: ETH-SLM frames are marked as drop ineligible to ensure accurate measurement.

A Maintenance Intermediate Point (MIP) does not process ETH-SLM frames, making it transparent to this measurement function.

2.4.1 Single-Ended Synthetic Loss Measurement

Single-ended ETH-SLM measures frame loss by exchanging synthetic packets between a local MEP and a Remote MEP (RMEP).

Procedure

1. The local MEP sends an SLM (Synthetic Loss Measurement) message to the RMEP.
2. The RMEP responds with an SLR (Synthetic Loss Reply) message, containing frame count information.
3. The local MEP calculates near-end and far-end frame loss based on transmitted and received frame counts.

Each SLM frame carries a TxFCI value representing the transmit counter at the time of sending, while the SLR frame includes:

- ✓ TxFCf: Transmit counter when the SLM was sent.
- ✓ TxFCb: Receive counter at the time of SLR transmission.

Using these values, the MEP calculates frame loss:

$$\text{Frame Loss}_{\text{far-end}} = |\text{TxFCf}[\text{tc}] - \text{TxFCf}[\text{tp}]| - |\text{TxFCb}[\text{tc}] - \text{TxFCb}[\text{tp}]|$$

$$\text{Frame Loss}_{\text{near-end}} = |\text{TxFCb}[\text{tc}] - \text{TxFCb}[\text{tp}]| - |\text{RxFCI}[\text{tc}] - \text{RxFCI}[\text{tp}]|$$

Where tc is the time of the current measurement and tp is the previous measurement time.

2.4.2 Dual-Ended Ethernet Synthetic Loss Measurement

Dual-ended ETH-SLM is a proactive or on-demand method for measuring frame loss in both point-to-point and multipoint Ethernet connections. It enables a Maintenance End Point (MEP) to periodically transmit ISL frames to its peer MEP(s) for frame loss measurement. The receiving MEP processes the frames and calculates near-end frame loss without forwarding them further.

Modes of Operation

- ✧ On-Demand: Initiated manually for diagnostic purposes.
- ✧ Proactive: Runs continuously to monitor ongoing network performance.

The selection between these modes is managed internally and does not need to be conveyed in the Protocol Data Unit (PDU).

1SL Frame Transmission

When configured for dual-ended ETH-SLM, a MEP transmits 1SL frames containing:

- ✓ Test ID: A unique identifier to distinguish multiple simultaneous tests.
- ✓ Source MEP ID: Identifies the transmitting MEP within the Maintenance Entity Group (MEG).
- ✓ TxFCf: The local frame transmit counter at the time of 1SL frame transmission.

The 1SL PDU is transmitted at a configured interval based on the performance monitoring requirements.

1SL Frame Reception & Frame Loss Calculation

Upon receiving a 1SL frame, the peer MEP verifies its validity by checking:

- MEG level
- Destination MAC address (matches its own or a multicast Class 1 MAC address)

The receiving MEP expects the next 1SL frame with an incremented TxFCf value. At the end of a measurement period, near-end frame loss is determined using:

$$\text{Frame Loss}_{\text{near-end}} = | \text{TxFCf}[\text{tc}] - \text{TxFCf}[\text{tp}] | - | \text{RxFCl}[\text{tc}] - \text{RxFCl}[\text{tp}] |$$

Where:

- ✓ TxFCf[tc]: Frame count in the last received 1SL frame at the end of the measurement period (tc).
RxFCl[tc]: Received frame count at the end of the measurement period (tc).
- ✓ TxFCf[tp]: Frame count in the first received 1SL frame at the start of the measurement period (tp).
- ✓ RxFCl[tp]: Received frame count at the start of the measurement period (tp).

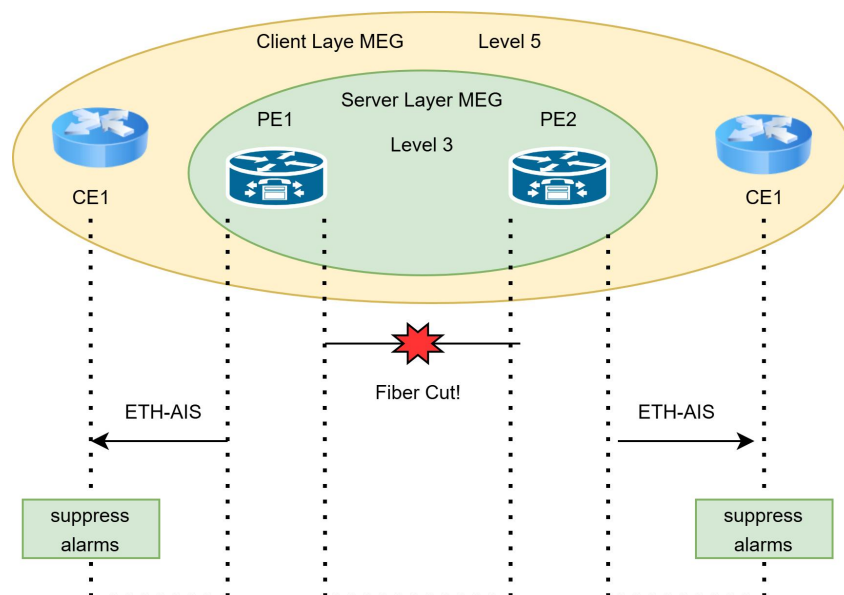
Dual-ended ETH-SLM is a crucial tool in carrier-grade Ethernet networks, enabling effective proactive monitoring and troubleshooting.

3. Fault Management

Fault management in ITU-T Y.1731 extends the capabilities of IEEE 802.1ag (CFM) by providing additional mechanisms for detecting, isolating, and notifying faults in carrier-grade Ethernet networks. While Continuity Check (CCM), Loopback (LB), and Link Trace (LT), as defined in 802.1ag CFM, are fundamental for connectivity verification and fault localization (already discussed in CFM), Y.1731 introduces advanced OAM functions to enhance fault detection and service reliability.

3.1 Ethernet Alarm Indication Signal (ETH-AIS)

The Ethernet Alarm Indication Signal (ETH-AIS) function in ITU-T Y.1731 is used to suppress unnecessary alarms when a defect condition is detected at a lower-layer network (server layer). This prevents multiple alarm notifications for the same failure across different layers, improving network stability and fault management. ETH-AIS is particularly useful in environments without independent restoration mechanisms, such as those not using Spanning Tree Protocol (STP).



ETH-AIS Transmission

- ✓ A MEP (Maintenance End Point), including a server MEP, generates AIS frames upon detecting a signal failure (e.g., loss of continuity in ETH-CC or an upstream AIS/LCK condition).
- ✓ The AIS frames are transmitted periodically to the client MEG level to notify upper-layer OAM entities to suppress alarms.
- ✓ Transmission continues until the defect condition is resolved.
- ✓ The first AIS frame must be transmitted immediately after detecting a defect, with a recommended transmission period of 1 second.

ETH-AIS Reception

- ✓ When a MEP receives an AIS frame, it examines whether the MEG level matches its own and detects the AIS defect condition.
- ✓ Alarm suppression occurs for all peer MEPs since the AIS frame does not specify a particular failed link.
- ✓ If no AIS frames are received within 3.5 times the AIS transmission period, the MEP clears the AIS defect condition and resumes alarm reporting for continuity loss.

3.2 Ethernet Test Signal (ETH-Test)

The Ethernet Test Signal (ETH-Test) function in ITU-T Y.1731 is used to perform one-way, on-demand diagnostics tests to assess throughput, frame loss, and bit errors. ETH-Test can be conducted in either in-service or out-of-service modes, ensuring flexible performance evaluation without impacting live traffic unless necessary.

Out-of-Service ETH-Test:

- ✓ Temporarily disrupts client traffic for diagnostics.
- ✓ The initiating MEP transmits ETH-LCK frames to notify downstream network elements about the test.

In-Service ETH-Test:

- ✓ Runs diagnostics without affecting active traffic.
- ✓ ETH-Test frames are transmitted at a controlled rate within a predefined portion of the bandwidth.

ETH-Test Transmission & Reception

- ✓ A MEP generates and inserts ETH-Test frames (TST frames) with a configurable throughput, frame size, and test pattern.
- ✓ The peer MEP receives the frames, analyzes them, and performs the necessary performance measurements.
- ✓ MIPs are transparent to ETH-Test frames and do not process them.

ETH-Test Functions

The ETH-Test function supports testing and diagnosing service interruptions or performance issues. It includes the following tests:

1. Verifying Throughput and Packet Loss: Throughput is measured by sending test frames at an increasing rate until frames are dropped.

Steps:

- ✓ Send frames at a specified rate and collect the number of sent and received packets.
- ✓ If packet loss occurs (sent > received), reduce the transmission rate.
- ✓ Repeat the process until the transmission rate results in no packet loss.
- ✓ The rate at which frames are transmitted without loss is the throughput for the link.

2. Verifying Bit Errors: Test frames carry a Cyclic Redundancy Check (CRC) code in the Test TLV field to detect bit errors.

Steps:

- ✓ Frames are transmitted with a specific pattern (e.g., Null signal or Pseudo Random Binary Sequence - PRBS).
- ✓ The receiver recalculates the CRC and compares it with the received CRC.
- ✓ If the codes match, there is no error; otherwise, a bit error is detected.

ETH-Test is a crucial performance monitoring tool in carrier-grade Ethernet networks, ensuring reliable service quality and SLA compliance.

3.3 Ethernet Locked Signal (ETH-LCK)

The Ethernet Locked Signal (ETH-LCK) function in ITU-T Y.1731 is used to indicate the administrative locking of a server-layer MEP, leading to a temporary suspension of data traffic towards the affected MEP. ETH-LCK helps differentiate between a planned maintenance action and an actual network defect. This function is crucial for out-of-service ETH-Test, where traffic is deliberately halted for diagnostics while preventing unnecessary alarms in upper-layer OAM systems.

ETH-LCK Transmission

- ✓ A server-layer MEP transmits ETH-LCK frames to its client-layer MEGs to notify them of an administrative lock.
- ✓ The first LCK frame must be sent immediately after the administrative action is initiated.
- ✓ The LCK transmission period is the same as AIS transmission (typically 1 second).
- ✓ If multiple client-layer MEGs exist, all must receive the first ETH-LCK frame within 1 second of the maintenance action.
- ✓ ETH-LCK transmission continues until the administrative condition is removed.

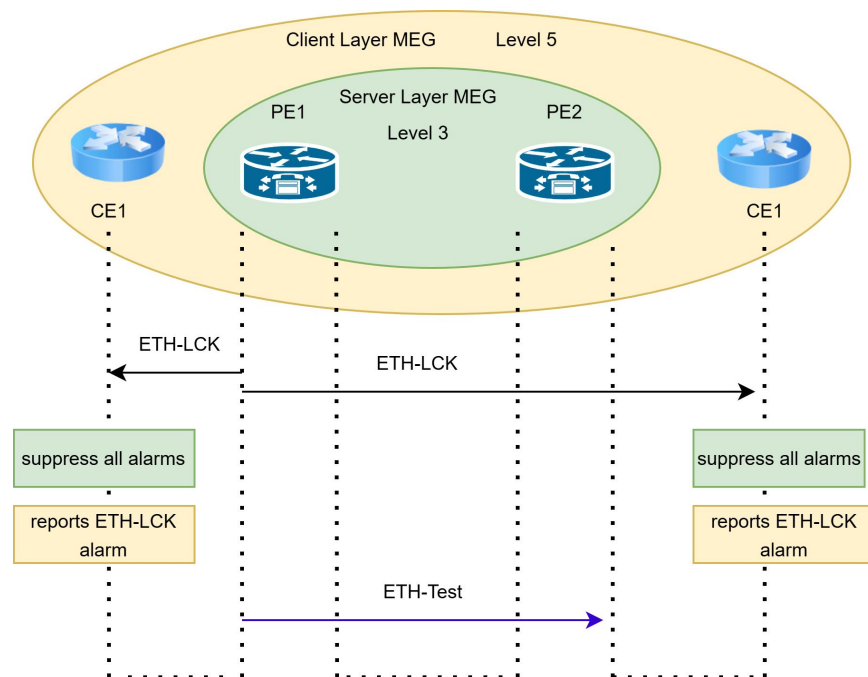
ETH-LCK in Multi-Domain Networks

In hierarchical maintenance levels, ETH-LCK helps outer MEPs distinguish between:

1. Fault-triggered alarms due to network failures (AIS-based alarm suppression).
2. Administrative lock-triggered alarms during planned maintenance (LCK-based alarm suppression).

For example, in a multi-level MEG environment:

- ✓ A lower-level MEP (inner MD) initiates an out-of-service ETH-Test and sends ETH-LCK to a higher-level MEP (outer MD).
- ✓ Upon receiving ETH-LCK, the outer MEP suppresses alarms and reports an ETH-LCK alarm.
- ✓ The inner MEP continues sending ETH-LCK frames throughout the test duration.
- ✓ Once the test completes, ETH-LCK frames stop, and if the outer MEP does not receive ETH-LCK within 3.5 times the transmission period, it resumes normal alarm reporting.



ETH-LCK Reception & Alarm Suppression

- ✓ A receiving MEP processes LCK frames at its configured MEG level and detects an LCK condition.
- ✓ This LCK condition triggers a signal fail state, which may cause the transmission of AIS (Alarm Indication Signal) frames to suppress unnecessary alarms in higher OAM layers.
- ✓ If no ETH-LCK frames are received within 3.5 times the configured transmission interval, the MEP clears the LCK condition and resumes normal operations.

Maintenance Intermediate Points (MIPs) are transparent to ETH-LCK frames and do not process or forward them.

3.4 Ethernet Client Signal Fail (ETH-CSF)

The Ethernet Client Signal Fail (ETH-CSF) function in ITU-T Y.1731 enables a Maintenance End Point (MEP) to notify its peer MEP of a detected failure in an Ethernet client signal. ETH-CSF is particularly useful when the client network does not support built-in fault detection mechanisms like Continuity Check (ETH-CC) or Alarm Indication Signal (ETH-AIS).

Scope & Applicability

- ✓ ETH-CSF is only applicable to point-to-point Ethernet transport applications.
- ✓ It must not be used in Ethernet switching environments like IEEE 802.1Q VLAN-based networks, except for point-to-point Ethernet flows.
- ✓ ETH-CSF is used to indicate client-layer failures as described in TU-T G.806.

ETH-CSF Transmission

- ✓ A MEP detects a client failure at its ingress port and begins transmitting ETH-CSF frames to its peer MEP.
- ✓ Transmission continues periodically until the failure condition is cleared.
- ✓ Clearing ETH-CSF can occur in two ways:
 1. Stopping ETH-CSF transmission when the failure is resolved.
 2. Sending a Client Defect Clear Indication (C-DCI) frame to explicitly notify the remote MEP that the failure has been cleared.

ETH-CSF Reception

- ✓ A receiving MEP detects an Ethernet remote CSF condition upon receiving a CSF frame.
- ✓ The MEP propagates the defect condition towards its egress client port to inform the client of the detected failure.
- ✓ A CSF condition is cleared when:
 1. No ETH-CSF frames are received within 3.5 times the configured CSF transmission interval.
 2. A CSF PDU with C-DCI (Client Defect Clear Indication) is received.

Types of ETH-CSF Defects

ETH-CSF messages include the type of defect detected, categorized as:

- 1) Client Loss of Signal (C-LOS) – Indicates a loss of the client signal.
- 2) Client Forward Defect Indication (C-FDI) – Notifies of a detected failure in the forward direction.
- 3) Client Reverse Defect Indication (C-RDI) – Notifies of a defect in the reverse direction.

Maintenance Intermediate Points (MIPs) are transparent to ETH-CSF frames and do not process or forward them.

ETH-CSF is essential for carrier-grade Ethernet networks, ensuring end-to-end fault visibility and enhanced network reliability.

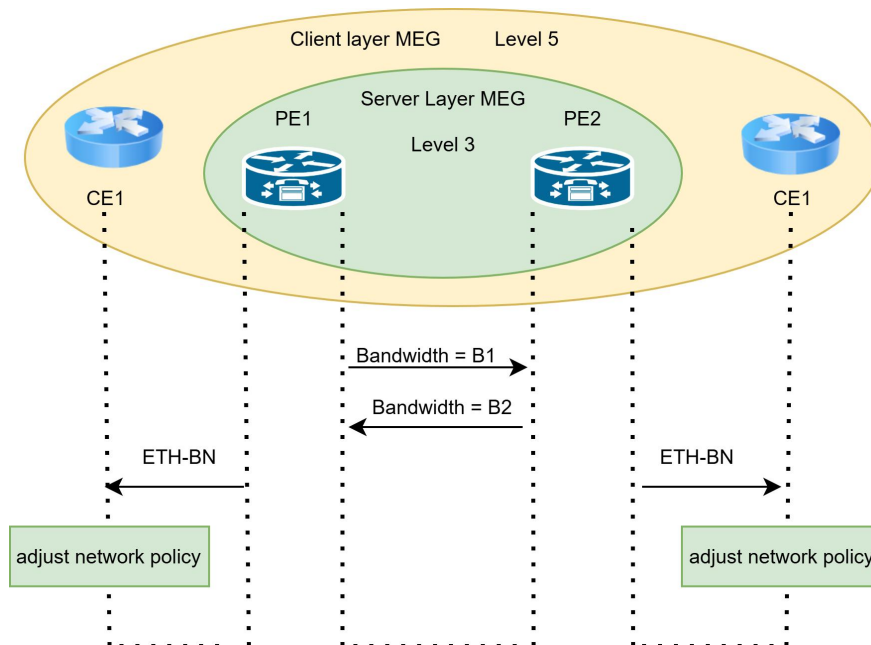
3.5 Ethernet Bandwidth Notification (ETH-BN)

The Ethernet Bandwidth Notification (ETH-BN) function in ITU-T Y.1731 enables server-layer MEPs to inform client-layer MEPs of changes in available bandwidth. This is particularly useful in microwave transport networks, where bandwidth can dynamically fluctuate due to environmental conditions.

ETH-BN helps client devices (e.g., routers) adjust traffic policies in response to bandwidth changes in the server layer, optimizing network performance and preventing congestion.

ETH-BN Transmission

- ✓ A server-layer MEP detects a bandwidth degradation and transmits periodic ETH-BN frames to inform its peer server-layer MEP about the reduced bandwidth.
- ✓ The first ETH-BN frames after a bandwidth change are sent in quick succession to ensure reliable notification, even if some frames are lost.
- ✓ ETH-BN frames may also be sent even when no degradation is present, providing continuous bandwidth status updates.
- ✓ If a link failure is detected, ETH-BN may indicate that the bandwidth has dropped to zero, and ETH-AIS (Alarm Indication Signal) frames may also be sent.



Hold Time for Bandwidth Change Notification

- ✓ A hold time (up to 10s) is applied before sending ETH-BN frames when a bandwidth change is detected.
- ✓ If the bandwidth change lasts for less than the hold time, no ETH-BN frame is sent, avoiding unnecessary notifications for short-term fluctuations (e.g., interference in a microwave link).

ETH-BN Reception

- ✓ A receiving MEP processes the ETH-BN frames.
- ✓ The management system can then use this data for network policy adjustments.
- ✓ If no ETH-BN frames are received within 3.5 times the last ETH-BN transmission interval, the receiving MEP assumes that bandwidth has been restored to full capacity and clears the degraded bandwidth condition.

ETH-BN in a Multi-Layer Network

In a multi-layer Ethernet network, ETH-BN allows server-layer MEPs (e.g., microwave devices) to notify client-layer MEPs (e.g., routing devices) about bandwidth changes.

- ✓ When server-layer bandwidth decreases, ETH-BN frames are sent to client-layer devices, which can adjust service policies accordingly (e.g., reduce traffic flow to degraded links).
- ✓ If the server-layer bandwidth recovers, ETH-BN transmission may stop based on configuration, and the client-layer devices age out the ETH-BN data after 3.5 times the last transmission interval.

ETH-BN is essential for carrier-grade Ethernet networks, ensuring efficient bandwidth management and optimized traffic flow in dynamically changing environments like microwave transport networks.

4. Application Scenario for Fault Management

4.1 Scenario 1: Service Disruption in a Multi-Layer Carrier Ethernet Network

In a Carrier Ethernet network, different layers handle transport and service functions. If a transport-layer failure occurs, it can trigger unnecessary alarms at the service layer. ETH-AIS, ETH-LCK, and ETH-RDI work together to ensure proper fault propagation and alarm suppression.

4.1.1 Network Setup

- ✓ A service provider operates a Metro Ethernet network with MPLS-based aggregation.
- ✓ Provider Edge (PE) routers are connected through transport switches, forming the server layer.
- ✓ Customers use Ethernet VPN (EVC) services that rely on service-layer MEPs.

4.1.2 Step-by-Step Fault Scenario

Step 1: Transport Layer Failure Occurs

- ✓ A fiber cut happens between two transport switches, disrupting traffic between PE1 and PE2.
- ✓ The server-layer MEP detects the signal fail condition (SF) and triggers a fault management response.

Step 2: ETH-AIS Propagation to Client Layer (Alarm Suppression)

- ✓ The server-layer MEP transmits ETH-AIS frames downstream to notify client-layer MEPs.
- ✓ The client-layer MEPs suppress unnecessary continuity check alarms, preventing alarm flooding.

Step 3: ETH-RDI Notifies the Remote MEP (Defect Indication)

- ✓ The affected service-layer MEP detects loss of continuity and sets the RDI flag in CCM messages.
- ✓ The remote MEP (on the other side of the failure) receives ETH-RDI, confirming the fault at the peer end.

Step 4: ETH-LCK Used for Maintenance (Planned Service Locking)

- ✓ The provider decides to perform maintenance on the failed link.
- ✓ The service-layer MEPs send ETH-LCK frames to notify all endpoints that this link is under administrative lock.
- ✓ Traffic is rerouted to alternative paths without triggering alarms.

Step 5: Recovery & Alarm Clearance

- ✓ The fiber cut is repaired, and the server-layer MEP stops sending ETH-AIS frames.
- ✓ The client-layer MEPs wait for 3.5 times the AIS transmission interval before resuming normal alarm reporting.
- ✓ The ETH-RDI flag is cleared in CCM messages once continuity is restored.
- ✓ The ETH-LCK condition is removed, and normal traffic flow resumes.

4.2 Scenario 2: Bandwidth Degradation in a Microwave Transport Network

In microwave-based transport networks, bandwidth is dynamically adjusted based on weather conditions. When bandwidth degrades, client-layer devices (e.g., routers) must adapt traffic accordingly. ETH-BN and ETH-CSF enable seamless bandwidth adaptation.

4.2.1 Network Setup

- ✓ Microwave links connect two transport devices (Device 2 & Device 3).
- ✓ Client-layer routers (Device 1 & Device 4) rely on these microwave links for service traffic.
- ✓ Server-layer MEPs monitor the bandwidth of microwave links.

4.2.2 Step-by-Step Bandwidth Degradation Scenario

Step 1: Microwave Link Experiences Bandwidth Reduction

- ✓ Due to rain fade, the microwave link between Device 2 and Device 3 reduces bandwidth.
- ✓ The server-layer MEP detects a bandwidth degradation event.

Step 2: ETH-BN Notifies Client Devices (Bandwidth Change Propagation)

- ✓ The server-layer MEPs send ETH-BN frames to client-layer routers (Device 1 & Device 4).
- ✓ ETH-BN frames contain bandwidth change information, allowing client devices to adjust accordingly.

Step 3: Client Devices Adjust Traffic Policies

- ✓ The client-layer routers use ETH-BN information to apply traffic shaping.
- ✓ Low-priority traffic (e.g., non-critical data) is rate-limited to ensure higher-priority services continue.

Step 4: Bandwidth Restores & ETH-BN Frames Stop

- ✓ Once weather conditions improve, the microwave link restores bandwidth.
- ✓ The server-layer MEP stops sending ETH-BN frames, signaling that bandwidth is back to normal.
- ✓ The client-layer devices remove bandwidth restrictions and restore normal traffic flow.

Handling Client Signal Failures (ETH-CSF)

- ✓ If the microwave link completely fails, the server-layer MEP detects client signal fail (CSF).
- ✓ ETH-CSF frames are transmitted to notify the peer MEP that the client signal is lost.
- ✓ The peer MEP propagates the failure to the client-layer routers, triggering backup path selection.

5. Conclusion

The ITU-T Y.1731 standard is a comprehensive framework for Operations, Administration, and Maintenance (OAM) in Carrier Ethernet networks, enabling both fault management and performance monitoring to ensure high service quality and reliability.

Key Takeaways:

Fault Management Functions (ETH-AIS, ETH-LCK, ETH-RDI, ETH-BN, ETH-CSF)

- ✓ Detect and localize faults in Ethernet networks.
- ✓ Suppress unnecessary alarms and improve network stability.
- ✓ Facilitate proactive fault recovery and maintenance.

Performance Monitoring Functions (ETH-LM, ETH-DM, ETH-SLM)

- ✓ Measure packet loss, delay, and synthetic loss in real time.
- ✓ Ensure networks meet Service Level Agreements (SLAs).
- ✓ Optimize quality for VoIP, video streaming, and critical enterprise applications.

With the increasing demand for high-performance and resilient Ethernet services, Y.1731 plays a crucial role in ensuring network reliability, optimizing performance, and reducing downtime. By implementing fault detection, latency monitoring, and loss measurement, service providers can deliver carrier-grade Ethernet services with minimal disruptions and maximum efficiency.

As next-generation networks evolve with 5G, cloud computing, and SDN/NFV, the importance of real-time OAM mechanisms like Y.1731 will only grow. Mastering these techniques will be essential for network engineers and service providers in maintaining scalable and high-quality Ethernet networks.

Y.1731 is not just a protocol—it's the backbone of a high-performance, fault-resilient Ethernet network!